

In preparation for designing a solvent system for a chemical reaction, a scientist reviewed the measurements of the solubility of ammonium chloride (NH_4Cl) reported in the scientific literature for four common organic amide solvents. The various reported values (expressed in three different units of concentration) were averaged, yielding the results in Table 1.

Table 1 Solubility of NH_4Cl in Selected Amide Solvents at 25 °C

Solvent		Solubility of NH_4Cl		
Name	Density (g/mL)	g/100 g of solvent	g/L	Molarity
Formamide	1.13	10.95	124.2	2.32
<i>N</i> -methylformamide	1.00	5.89	*	1.10
<i>N,N</i> -dimethylformamide	0.90	0.17	1.6	0.03
<i>N</i> -methylacetamide	0.96	*	*	0.90
* Note: Value not reported in the literature examined				

If a 1.40 g sample of ammonium chloride is placed in 25.0 mL of *N*-methylformamide at 25 °C, the sample will:

- ✔ ☒ A. fully dissolve to form an unsaturated solution, because the amount of the sample will not exceed its solubility limit. [58%]
- ☐ B. fully dissolve to form a saturated solution, because the amount of the sample will exactly reach its solubility limit. [15%]
- ☐ C. partially dissolve to form a saturated solution, because the amount of the sample will exceed its solubility limit. [23%]
- ☐ D. not dissolve at all, because the sample is completely insoluble in the solvent. [2%]

Correct

57% answered correctly

Explanation:

Sample fully dissolves when solute-solvent ratio is less than the solubility limit

1.40 g NH_4Cl

25.0 mL

N-methylformamide (density = 1.0 g/mL)

Mass ratio: $\frac{1.40 \text{ g } \text{NH}_4\text{Cl}}{25.0 \text{ g solvent}}$

Solubility limit: $\frac{5.89 \text{ g } \text{NH}_4\text{Cl}}{100. \text{ g solvent}}$

Sample fully dissolves

Solubility refers to the maximum amount of a **solute** (ie, a dissolved compound) that can dissolve in a given amount of a **solvent** (ie, the solution medium) at a given temperature.

Solubilities are sometimes expressed in terms of mass per unit volume (eg, g/L, mg/mL) or in terms of the moles per unit volume (eg, **molar solubility** = mol/L). However, the density (and

therefore the volume) of a solvent varies with temperature. In scientific literature, solubility measurements are commonly reported as a **mass ratio** (ie, the **grams of solute** that will dissolve per **100 g of solvent**) because mass does not vary with temperature.

Although the mass of the substances does not vary with temperature, increasing the temperature generally increases the **amount of solute** that can dissolve. As a result, reported solubility measurements will always state the temperature at which the measurement was made. Most commonly, temperatures of 20 or 25 °C are used as convenient values for "room temperature" comparisons.

If a 1.40 g sample of NH_4Cl is placed in 25.0 mL of *N*-methylformamide at 25 °C, this yields a mass-to-volume ratio of:

$$\frac{1.40 \text{ g NH}_4\text{Cl solute}}{25.0 \text{ mL solvent}}$$

Although Table 1 does not give the solubility in units of g/mL, this ratio can be compared to the given data by applying the solvent density. Because the density of *N*-methylformamide is 1.0 g/mL at 25 °C, the mass-to-mass ratio is numerically equivalent.

$$\frac{1.40 \text{ g NH}_4\text{Cl solute}}{25.0 \text{ mL solvent}} \times \frac{1 \text{ mL solvent}}{1.0 \text{ g solvent}} = \frac{1.40 \text{ g NH}_4\text{Cl solute}}{25.0 \text{ g solvent}}$$

To easily compare this ratio to the mass ratio solubility given in Table 1, the ratio can be scaled by a factor of 4 (which multiplies the ratio by 1 because $4/4 = 1$).

$$\frac{1.40 \text{ g NH}_4\text{Cl solute}}{25.0 \text{ g solvent}} \times \frac{4}{4} = \frac{5.60 \text{ g NH}_4\text{Cl solute}}{100. \text{ g solvent}}$$

The proportionately scaled result is less than the maximum solubility of NH_4Cl in *N*-methylformamide reported in Table 1.

$$\frac{5.60 \text{ g NH}_4\text{Cl solute}}{100. \text{ g solvent}} < \frac{5.89 \text{ g NH}_4\text{Cl solute}}{100. \text{ g solvent}}$$

Therefore, 1.40 g of NH_4Cl placed in 25.0 mL of *N*-methylformamide at 25 °C will fully dissolve, because the amount of the sample does not exceed its solubility limit.

(Choices B and C) The mass ratio of the sample in the solvent does not equal or exceed the reported solubility.

(Choice D) Table 1 indicates that NH_4Cl is partially soluble in *N*-methylformamide.

Educational objective:

Solubility refers to the maximum amount of a solute that can dissolve in a given amount of a solvent at a given temperature. Solubility measurements are usually reported as the grams of solute that will dissolve per 100 g of solvent.

Time spent: 0 sec

Subject:
General Chemistry

QID: 403910

System:
Solutions and Electrochemistry

In preparation for designing a solvent system for a chemical reaction, a scientist reviewed the measurements of the solubility of ammonium chloride (NH₄Cl) reported in the scientific literature for four common organic amide solvents. The various reported values (expressed in three different units of concentration) were averaged, yielding the results in Table 1.

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Solvent		Solubility of NH ₄ Cl		
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Formamide	1.13	10.95	124.2	2.32
<i>N</i> -methylformamide	1.00	5.89	*	1.10
<i>N,N</i> -dimethylformamide	0.90	0.17	1.6	0.03
<i>N</i> -methylacetamide	0.96	*	*	0.90

* Note: Value not reported in the literature examined

What is the solubility of NH₄Cl in *N*-methylacetamide expressed as grams NH₄Cl per 100 grams of solvent? (Note: The molar mass of NH₄Cl is 53.5 g/mol.)

- ☐ A. 0.050 g/100 g solvent [19%]
☐ B. 0.094 g/100 g solvent [12%]
☒ C. 5.0 g/100 g solvent [55%]
☐ D. 5.9 g/100 g solvent [12%]

Correct

54% answered correctly

Explanation:

Conversion of solubility in alternate units

48.2 g NH₄Cl (0.90 mol)

960 g solvent *N*-methylacetamide (density = 0.96 g/mL)

1.0 L

Mass ratio per gram of solvent

$$\frac{48.2 \text{ g NH}_4\text{Cl}}{960 \text{ g solvent}} = \frac{0.050 \text{ g NH}_4\text{Cl}}{\text{g solvent}}$$

Ratio per 100 g

$$\frac{0.050 \text{ g NH}_4\text{Cl}}{\text{g solvent}} \times \frac{100}{1} = \frac{5.0 \text{ g NH}_4\text{Cl}}{100 \text{ g solvent}}$$

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Solubility refers to the maximum amount of a **solute** (ie, a dissolved compound) that can dissolve in a given amount of a **solvent** (ie, the solution medium) at a given temperature. Solubility measurements are commonly reported as a **mass ratio** of the **grams of solute** that will dissolve per **100 g of solvent** because, unlike solvent density and volume, mass does not vary with temperature.

However, when performing reactions in lab settings, it is often convenient to express solubilities in terms of mass per unit volume (eg, g/L, mg/mL) or in terms of the moles per unit volume (eg, **molar solubility** = mol/L). If the solubility is known in these units, the measurement can be converted to a mass ratio using the molar mass of the solute and the solution density. For *dilute* solutions, the solution density is approximately equal to the solvent density.

In Table 1, the solubility of NH₄Cl in *N*-methylacetamide is not reported in grams per 100 g of solvent, but it is reported in **molarity** (mol/L). Multiplying the molar solubility by the molar mass of NH₄Cl yields the solubility in g/L.

$$\frac{0.90 \text{ mol NH}_4\text{Cl solute}}{\text{L solution}} \times \frac{53.5 \text{ g NH}_4\text{Cl solute}}{1 \text{ mol NH}_4\text{Cl solute}} = \frac{48.2 \text{ g NH}_4\text{Cl solute}}{\text{L solution}}$$

Converting the result to units of grams per milliliter and dividing by the solvent density (as an approximation of the solution density) yields the solubility as a mass ratio of grams solute per *gram* of solvent.

$$\frac{48.2 \text{ g NH}_4\text{Cl solute}}{\text{L solution}} \times \frac{1 \text{ L solution}}{1000 \text{ mL solution}} = \frac{0.0482 \text{ g NH}_4\text{Cl solute}}{\text{mL solution}}$$
$$\frac{0.0482 \text{ g NH}_4\text{Cl solute}}{\text{mL solution}} \times \frac{1 \text{ mL solution}}{0.96 \text{ g solvent}} \approx \frac{0.050 \text{ g NH}_4\text{Cl solute}}{1 \text{ g solvent}}$$

To express the mass-to-mass solubility ratio per *100 g of solvent*, the ratio must be scaled (multiplied) by a factor of 100. The relative ratio is unchanged because multiplying by 100/100 is the same as multiplying by 1 (ie, 100/100 = 1).

$$\frac{0.050 \text{ g NH}_4\text{Cl solute}}{1 \text{ g solvent}} \times \frac{100}{100} = \frac{5.0 \text{ g NH}_4\text{Cl solute}}{100 \text{ g solvent}}$$

(Choice A) A mass of 0.050 g NH₄Cl dissolves per *gram* of solvent (not per 100 g of solvent).

(Choice B) A value of 0.094 (with incorrect units) results from failing to convert the moles of NH₄Cl to grams during the calculation.

(Choice D) NH₄Cl has a solubility of 5.9 g/100 g in *N*-methylformamide, but the question asks about the solubility in *N*-methylacetamide.

Educational objective:

A compound's molar solubility (ie, moles dissolved per liter of solution) can be converted to a mass ratio (ie, grams of solute per 100 g of solvent) using the solute's molar mass and the solution's density.

Time spent: 0 sec
Subject:
General Chemistry

QID: 403911
System:
Solutions and Electrochemistry

In preparation for designing a solvent system for a chemical reaction, a scientist reviewed the measurements of the solubility of ammonium chloride (NH_4Cl) reported in the scientific literature for four common organic amide solvents. The various reported values (expressed in three different units of concentration) were averaged, yielding the results in Table 1.

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Solvent		Solubility of NH_4Cl		
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<i>N,N</i> -dimethylformamide	0.90	0.17	1.6	0.03
<i>N</i> -methylacetamide	0.96	*	*	0.90

* Note: Value not reported in the literature examined

If the solubility of ammonium chloride dissolved in *N,N*-dimethylformamide is expressed in parts per million (ppm), what is the numerical value?

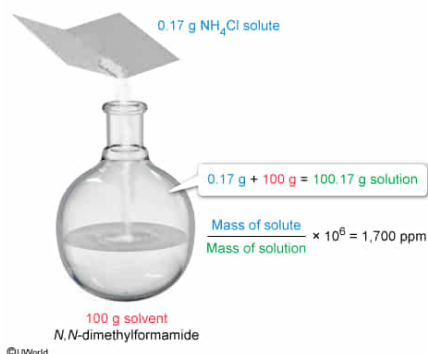
- ☐ A. 0.0017 [24%]
☐ B. 0.17 [6%]
☐ C. 1.7 [8%]
☒ D. 1,700 [60%]

Correct

60% answered correctly

Explanation:

Concentration expressed in parts per million (ppm)



Solubility measurements are usually reported as a **mass ratio** of the **grams of solute** that will dissolve per **100 g of solvent** because, unlike solvent density and volume, mass does not vary with temperature. A mass ratio can be alternatively expressed as a mass percent, which gives the number of units of solute mass per 100 units of solution mass.

$$\text{Mass percent} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100 = \frac{\text{Mass of solute}}{(\text{Mass of solute} + \text{Mass of solvent})} \times 100$$

However, a 100-unit basis (ie, parts per hundred) is often too small to conveniently express concentrations of very dilute solutions. In such cases, a one-million-unit basis (ie, $1,000,000 = 10^6$) is sometimes used to express the concentration in **parts per million (ppm)**, which is analogously defined as:

$$\text{ppm} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6 = \frac{\text{Mass of solute}}{(\text{Mass of solute} + \text{Mass of solvent})} \times 10^6$$

An approximation and two useful equivalences for ppm are worth mentioning:

- A **convenient approximation** is to divide the mass of the solute by the mass of the solvent (ie, the majority

contributor to the solution mass) when the mass of the solute is very small compared to the mass of the solvent.

$$\text{ppm} = \frac{\text{Mass of solute}}{(\text{Mass of solute} + \text{Mass of solvent})} \times 10^6 \approx \frac{\text{Mass of solute}}{\text{Mass of solvent}} \times 10^6$$

Caution: It is important to recognize the correct setup (if required by a question) and to be aware that the small numerical error introduced by the approximation will increase as the solute concentration increases.

- **ppm is approximately equivalent to mg/L** when the solution density is near 1.0 g/mL (eg, dilute solutions in water):

$$\text{ppm} \approx \frac{\text{Milligrams of solute}}{\text{Liter of solution}} = \frac{\text{mg}}{\text{L}}$$

- **ppm can be expressed as mg/kg** when dealing with large amounts of mass:

$$\text{ppm} = \frac{\text{Milligrams of solute}}{\text{Kilogram of solution}} = \frac{\text{mg}}{\text{kg}}$$

In Table 1, the solubility of NH_4Cl dissolved in *N,N*-dimethylformamide is given as 0.17 g/100 g solvent. As such, the equivalent concentration expressed in ppm is given by:

$$\begin{aligned}\text{ppm} &= \frac{\text{Mass of solute}}{(\text{Mass of solute} + \text{Mass of solvent})} \times 10^6 \\ \text{ppm} &= \frac{0.17\text{g solute}}{(0.17\text{g solute} + 100\text{g solvent})} \times 10^6 = \frac{0.17\text{g solute}}{100.17\text{g solution}} \times 10^6\end{aligned}$$

Applying a mathematical approximation (ie, 100.17 g $\approx$ 100 g) gives:

$$\text{ppm} = \frac{0.17\text{ g solute}}{100.17\text{ g solution}} \times 10^6 \approx \frac{0.17\text{ g solute}}{100.\text{g solution}} \times 10^6 = 0.0017 \times 10^6 = 1,700$$

(Choice A) A value of 0.0017 is simply the mass fraction of the solute in the solution:

$$\frac{0.17\text{ g solute}}{(0.17\text{ g solute} + 100\text{ g solvent})} \approx \frac{0.17\text{ g solute}}{100\text{ g solution}} = 0.0017$$

(Choice B) A value of 0.17 is the parts per hundred (ie, percent) of the solute in the solution:

$$\frac{0.17\text{ g solute}}{100.17\text{ g solution}} \times 100 \approx \frac{0.17\text{ g solute}}{100\text{ g solution}} \times 100 = 0.17$$

(Choice C) A value of 1.7 is the parts per thousand (ppth) of the solute in the solution:

$$\frac{0.17\text{ g solute}}{100.17\text{ g solution}} \times 1000 \approx \frac{0.17\text{ g solute}}{100\text{ g solution}} \times 1000 = 1.7$$

Educational objective:

Solubility measurements are usually reported as a mass ratio of the grams of solute that will dissolve per 100 g of solvent. However, for very dilute solutions, the ratio of the solute mass to the solution mass is sometimes expressed as parts per million (ppm), which is defined as:

$$\text{ppm} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6$$

Time spent: 0 sec
Subject:
General Chemistry

QID: 403912
System:
Solutions and Electrochemistry

In preparation for designing a solvent system for a chemical reaction, a scientist reviewed the measurements of the solubility of ammonium chloride (NH_4Cl) reported in the scientific literature for four common organic amide solvents. The various reported values (expressed in three different units of concentration) were averaged, yielding the results in Table 1.

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Name	Density (g/mL)	g/100 g of solvent	g/L	Molarity
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<i>N,N</i> -dimethylformamide	0.90	0.17	1.6	0.03
<i>N</i> -methylacetamide	0.96	*	*	0.90
* Note: Value not reported in the literature examined				

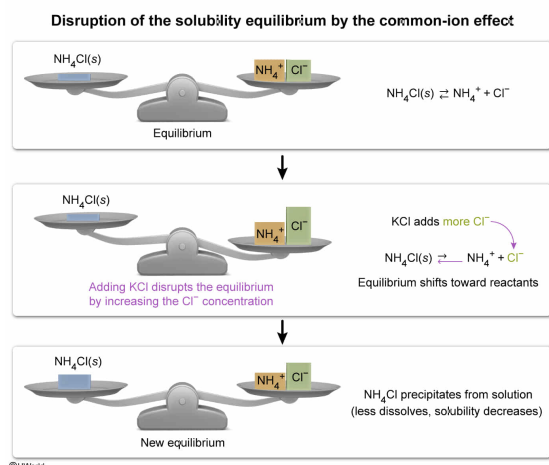
Compared to the solubility of the ammonium chloride in pure formamide, what effect would dissolving KCl into the formamide have on the solubility of ammonium chloride in the solvent?

- ☐ A. The solubility of the ammonium chloride would increase. [11%]
- ☐ B. The solubility of the ammonium chloride would be unchanged. [5%]
- ✓ ☒ C. The solubility of the ammonium chloride would decrease. [79%]
- ☐ D. The effect of the added KCl salt cannot be determined without more information. [2%]

Correct

79% answered correctly

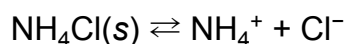
Explanation:



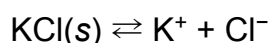
When an ionic compound is placed in water, the ions forming the compound will **dissociate** to some extent into the solution and establish an **equilibrium**. **Le Châtelier's principle** states that a disrupted equilibrium will shift to counteract the disruption.

Accordingly, if a solution contains two dissolved compounds that have an **ion in common**, the common ion supplied by one compound produces a **common-ion effect** that **shifts the equilibrium** of dissolution of the other compound to the **left** (ie, toward the undissolved state). Consequently, the common-ion effect decreases the amount of the compound that will dissolve (ie, lowers its solubility).

When dissolved in formamide, NH_4Cl dissociates into its component ions and establishes equilibrium:



Similarly, KCl dissolved in formamide dissociates into K^+ and Cl^- ions:



NH_4Cl and KCl have the Cl^- anion in common. Therefore, adding KCl to the formamide would produce a common-ion effect in which the added Cl^- ions would shift the NH_4Cl equilibrium to the left and lower the amount of NH_4Cl that would dissolve (ie, the solubility of the NH_4Cl would decrease).

(Choices A, B, and D) The effect of the added KCl can be predicted by Le Châtelier's principle, which indicates that the additional Cl^- ion from KCl will *decrease* the solubility of $\text{NH}_4\text{Cl}(s)$ by shifting the equilibrium toward undissolved solid due to the common-ion effect.

Educational objective:

In solutions containing two compounds that have an ion in common, the common ion supplied by one compound produces a common-ion effect that decreases the solubility of the other compound. This occurs by shifting the solubility equilibrium toward reactants according to Le Châtelier's principle.

Time spent:0 sec

Subject:
General Chemistry

QID:403913

System:
Solutions and Electrochemistry

In preparation for designing a solvent system for a chemical reaction, a scientist reviewed the measurements of the solubility of ammonium chloride (NH_4Cl) reported in the scientific literature for four common organic amide solvents. The various reported values (expressed in three different units of concentration) were averaged, yielding the results in Table 1.

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* Note: Value not reported in the literature examined

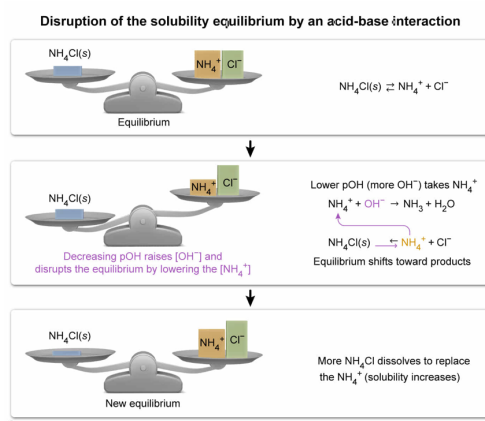
Which of the following modifications to the experimental procedure to prepare the solutions will most likely increase the amount of ammonium chloride that dissolves in the amide solvents in Table 1? (Note: The dissolution of ammonium chloride is endothermic.)

- ☐ A. Decreasing the pH of the solution [37%]
- ✓ ☒ B. Decreasing the pOH of the solution [38%]
- ☐ C. Adding a small amount of ammonium bromide salt [11%]
- ☐ D. Cooling the solution as the ammonium chloride is added [12%]

Correct

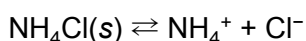
37% answered correctly

Explanation:



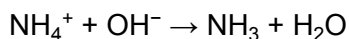
According to **Le Châtelier's principle**, if a reaction **equilibrium** is disrupted, it will shift to accommodate the disruption and establish a new equilibrium. This principle indicates that if some of the products of the reaction are removed, the equilibrium will shift to form more of the products.

When dissolved in formamide, NH_4Cl dissociates into its component ions and establishes equilibrium:



However, the NH_4^+ ion is a weak **acid**, and this property can be exploited to **disrupt the equilibrium**

because acids **react** with **bases** (eg, OH^-). The NH_4^+ ion reacts with OH^- according to the reaction:



Therefore, **decreasing the pOH** (ie, increasing the amount of OH^-) in the amide solvent solutions will cause some of the NH_4^+ to be consumed by the base, and the concentration of NH_4^+ will initially decrease. The decrease in NH_4^+ will disrupt the NH_4Cl equilibrium, causing a **shift toward the products** to form more NH_4^+ (ie, a greater amount of the compound will dissolve).

(Choice A) **Decreasing the pH** (ie, increasing the amount of H^+) will not cause an acid-base reaction to disrupt the solubility equilibrium because H^+ doesn't neutralize other acids (eg, NH_4^+) and Cl^- is not a good base.

(Choice C) Adding a small amount of NH_4Br salt will introduce NH_4^+ as a common ion, resulting in a **common-ion effect** that *decreases* the amount of NH_4Cl that will dissolve.

(Choice D) The solubility of ammonium chloride decreases as temperature decreases because the dissolution is endothermic (ie, requires some thermal energy).

Educational objective:

Changing the pH or pOH of a solution can affect the solubility of compounds that have an acidic or basic ion. This occurs by disrupting the solubility equilibrium according to Le Châtelier's principle through acid-base reactions with the ion.

Time spent:0 sec

Subject:
General Chemistry

QID:403914

System:
Solutions and Electrochemistry

In preparation for designing a solvent system for a chemical reaction, a scientist reviewed the measurements of the solubility of ammonium chloride (NH_4Cl) reported in the scientific literature for four common organic amide solvents. The various reported values (expressed in three different units of concentration) were averaged, yielding the results in Table 1.

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<i>N</i> -methylacetamide	0.96	*	*	0.90
* Note: Value not reported in the literature examined				

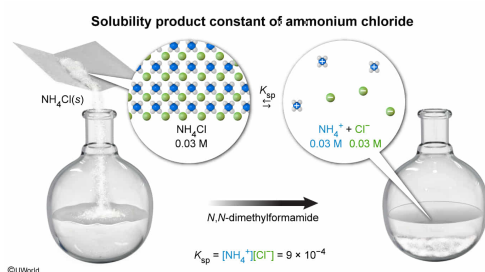
Based on the solubility data in Table 1, what is the solubility product constant K_{sp} for ammonium chloride in *N,N*-dimethylformamide?

- ✓ ☒ A. 9.0×10^{-4} [34%]
☐ B. 3.0×10^{-2} [43%]
☐ C. 6.0×10^{-2} [17%]
☐ D. 1.2×10^0 [4%]

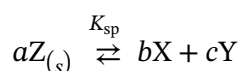
Correct

35% answered correctly

Explanation:



For an ionic compound Z that dissociates into ions X and Y when placed in a solvent, an equilibrium is established in which the rate of dissolution (forward reaction) equals the rate of precipitation (reverse reaction).



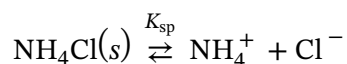
where a , b , and c are the **stoichiometric coefficients** of each component. This equilibrium position is quantified by the **solubility product constant** K_{sp} , which provides an indication of the solubility of a compound at a given temperature. The general K_{sp} expression is written as:

$$K_{\text{sp}} = [\text{X}]^b [\text{Y}]^c$$

where $[\text{X}]$ and $[\text{Y}]$ are the **molar concentrations** of the X and Y ions, and b and c are the respective

coefficients of the **balanced dissolution reaction**. The K_{sp} is the product of the maximum possible concentrations of each ion in the solution under the given conditions.

When dissolved in *N,N*-dimethylformamide, NH_4Cl dissociates into its component ions and establishes equilibrium:



As such, the K_{sp} expression for the equilibrium is:

$$K_{sp} = [\text{NH}_4^+][\text{Cl}^-]$$

According to Table 1, NH_4Cl dissolved in *N,N*-dimethylformamide has a **molar solubility** of 0.03 M at 25 °C. Because NH_4Cl dissociates into NH_4^+ and Cl^- ions in a 1:1:1 mole ratio, the molar concentrations of these species must also be 0.03 M.

$$[\text{NH}_4^+] = [\text{Cl}^-] = 0.03 \text{ M}$$

Substituting these concentrations into the K_{sp} expression and applying the **rules** for math operations with scientific notation gives:

$$K_{sp} = [\text{NH}_4^+][\text{Cl}^-] = (0.03)(0.03) = (3 \times 10^{-2})(3 \times 10^{-2}) = 9 \times 10^{-4}$$

(Choice B) A value of 3.0×10^{-2} is the molar concentration of the NH_4^+ and Cl^- ions, not the K_{sp} .

(Choice C) A value of 6.0×10^{-2} results from evaluating the K_{sp} as $2 \times (3.0 \times 10^{-2})$ rather than as $(3.0 \times 10^{-2}) \times (3.0 \times 10^{-2})$.

(Choice D) A value of 1.2×10^0 is the K_{sp} of NH_4Cl dissolved in *N*-methylformamide, but the question asks for the K_{sp} in *N,N*-dimethylformamide.

Educational objective:

The solubility product constant K_{sp} is the equilibrium constant for the dissolution of a compound in a saturated solution at a given temperature. K_{sp} is calculated as the product of the molar concentrations of the dissociated ion species raised to the power of their respective coefficients from the balanced dissolution reaction.

Time spent: 0 sec

Subject:
General Chemistry

QID: 403915

System:
Solutions and Electrochemistry